THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL UNIT TEST SET 2

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. $\sqrt{1.44}$ is equal to: 1

(a) 1.3 **(b)** 12.0 **(c)** 1.2 **(d)** 0.12

2. If $(2^3)^x = 2^{12}$, then x is: 1

(a) 15 **(b)** 9 **(c)** 4 **(d)** 36

3. The largest 3-digit number in base 2, written in decimal, is: 1

(a) 3 **(b)** 8 **(c)** 5 **(d)** 7

4. In a rhombus ABCD, $\angle A=(4x+10)^\circ$ and $\angle B=(5x+17)^\circ$. The value of x is: 1

(a) 21 **(b)** 18 **(c)** 17 **(d)** 16

5. The smallest number divisible by 4, 6 and 10 is: 1

(a) 30 **(b)** 64 **(c)** 60 **(d)** 120

6. $\sqrt[3]{0.027}$ is equal to: 1

(a) 3.0 **(b)** 0.3 **(c)** 0.4 **(d)** 0.03

7. $2^3 \times 5^3$ equals: 1

(a) 1000 **(b)** 30 **(c)** 250 **(d)** 133

8. The number of digits in the binary form of 20 is: 1

(a) 4 **(b)** 7 **(c)** 5 **(d)** 6

9. In a kite ABCD with AB = AD and CB = CD, $\angle A = 60^\circ$ and $\angle C = 130^\circ$. The angle $\angle B$ is: 1

(a) 85° **(b)** 170° **(c)** 95° **(d)** 75°

10. The smallest number to be added to 4829 to make it divisible by 8 is: 1

(a) 3 **(b)** 6 **(c)** 4 **(d)** 5

11. The value of 45^2 is: 1

(a) 2030 **(b)** 2050 **(c)** 2000 **(d)** 2025

12. The product $(2 \times 10^3) \times (3 \times 10^4)$ in scientific notation is: 1

(a) 6×10^8 **(b)** 6×10^7 **(c)** 6×10^6 **(d)** 60×10^7

13. $1011_2 + 110_2$ equals: 1

(a) 10011_2 **(b)** 10010_2 **(c)** 10001_2 **(d)** 10000_2

14. Three exterior angles of a quadrilateral are $80°$, $100°$ and $95°$. The fourth exterior angle is: 1

(a) $85°$ **(b)** $95°$ **(c)** $75°$ **(d)** $110°$

15. The smallest 4-digit number divisible by 7 is: 1

(a) 1000 **(b)** 1008 **(c)** 994 **(d)** 1001

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Evaluate $\sqrt[3]{\frac{27}{512}}$. 2

17. Find x if $2^x = \frac{1}{32}$. 2

18. Write CXXXIX + CCLXII as a Roman numeral. 2

19. The diagonals of a parallelogram ABCD meet at O with AO $= (3x + 5)$ cm and OC $= (2x + 9)$ cm. Find x and AC. 2

20. Check whether 3516 is divisible by 12 using the rules for 3 and 4. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Multiply $101_2 \times 11_2$ and give the answer in binary. 3

22. The adjacent sides of a parallelogram are in the ratio 2:5 and its perimeter is 56 cm. Find the lengths of its sides. 3

23. Find the digit * so that 83*5 is divisible by 11. Show your working. 3

24. Find $\sqrt{1.96}$ using prime factorisation of the whole number obtained after removing the decimal point. 3

OR

A factory makes 2.5×10^4 bolts each day. How many bolts does it make in 12 days? Give the answer in scientific notation. 3

25. Write 1994 and 449 in Roman numerals and find their difference in Roman numerals. 3

OR

In a rectangle ABCD, $\angle ACB = 25°$. Find $\angle CAB$, $\angle AOB$ and $\angle ODC$ where O is the point of intersection of the diagonals. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Apply the Kaprekar routine to 1495 until you reach 6174. Write every step and state the number of steps. 5

OR

The volume of a cube is 1728 cm³. Find the side of the cube and its total surface area. 5

27. The diameter of a cell is 2×10^{-5} cm. How many such cells placed side by side in a straight line make a length of 8×10^1 cm? Give the answer in scientific notation. 5

OR

Write the smallest and the largest 5-digit numbers in base 2 and find their decimal values. How many 5-digit numbers are there in base 2? 5

END OF QUESTION PAPER **Total: 50 Marks**

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